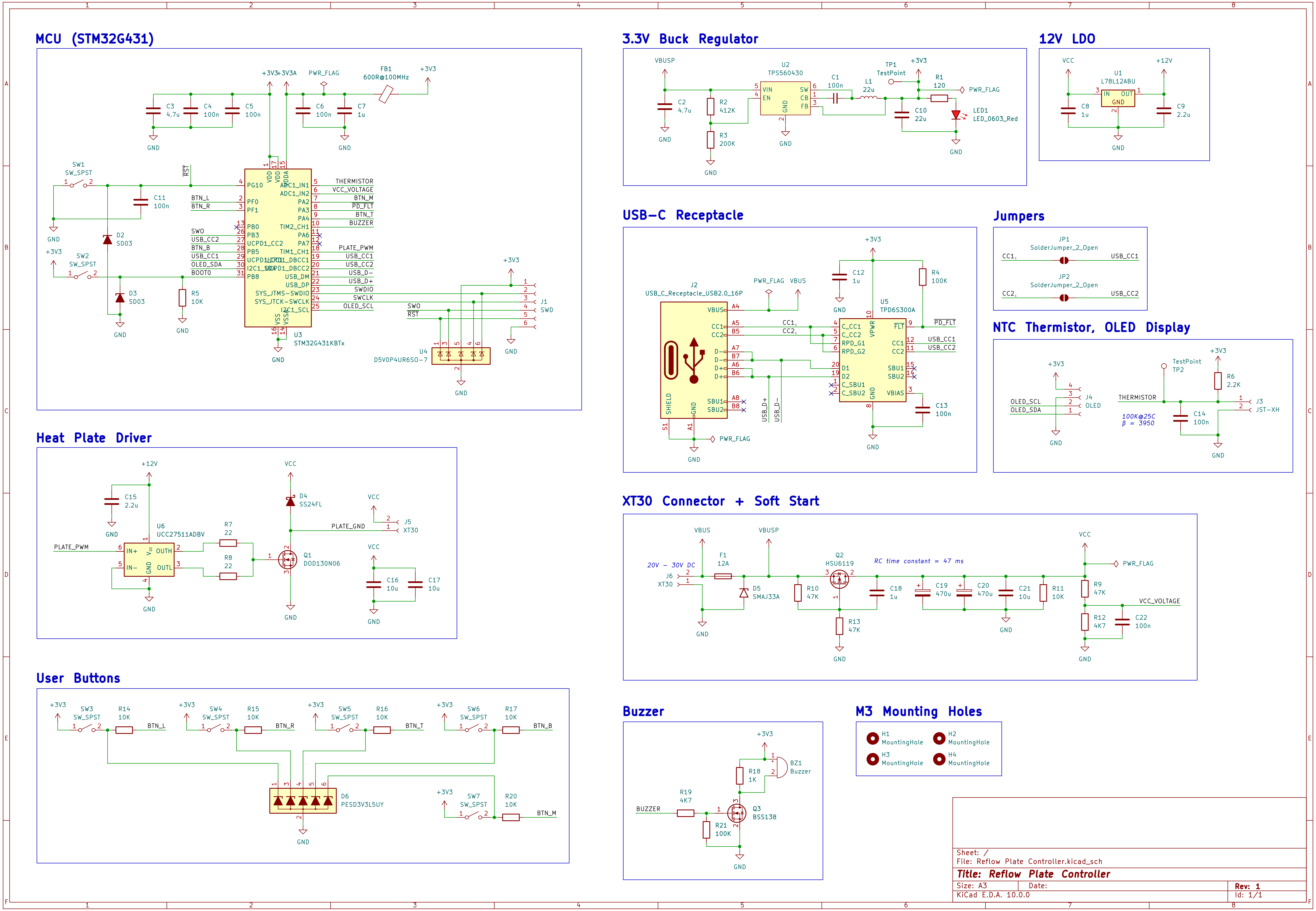

MCU (STM32G431)
+3V3 +3V3A PWR_FLAG
FB1 600R@100MHz
+3V3
C3 4.7u
C4 100n
C5 100n
C6 100n
C7 1u
GND
SW1 SW_SPST
RST
C11 100n
D2 SD03
SW2 SW_SPST
D3 SD03
R5 10K
U3 STM32G431KBTx
PG10 PF0 PF1 PB0 PB3 PB5 PB8
ADC1_IN1 ADC1_IN2 PA2 PA3 PA4 TIM2_CH1 PA6 PA7 TIM1_CH1 UCPD1_DBCC1 UCPD1_DBCC2 USB_DM USB_DP SYS_JTMS-SWDIO SYS_JTCK-SWCLK I2C1_SCL
THERMISTOR VCC_VOLTAGE BTN_M PD_FLT BTN_T BUZZER PLATE_PWM USB_CC1 USB_CC2 USB_D− USB_D+ SWDIO SWCLK OLED_SCL
BTN_L BTN_R SWO USB_CC2 BTN_B USB_CC1 OLED_SDA BOOT0
U4 D5V0P4UR6SO-7
J1 SWD
3.3V Buck Regulator
VBUSP
C2 4.7u
R2 412K
R3 200K
U2 TPS560430
VIN EN SW CB FB GND
C1 100n
L1 22u
TP1 TestPoint
+3V3
R1 120
PWR_FLAG
C10 22u
LED1 LED_0603_Red
12V LDO
VCC +12V
U1 L78L12ABU
IN OUT GND
C8 1u
C9 2.2u
USB−C Receptacle
J2 USB_C_Receptacle_USB2.0_16P
VBUS CC1 CC2 D− D+ SBU1 SBU2 SHIELD GND
PWR_FLAG VBUS
CC1 CC2
USB_D+ USB_D−
C12 1u
R4 100K
U5 TPD6S300A
C_CC1 C_CC2 RPD_G1 RPD_G2 D1 D2 C_SBU1 C_SBU2 VPWR FLT CC1 CC2 SBU1 SBU2 VBIAS
PD_FLT USB_CC1 USB_CC2
C13 100n
Jumpers
JP1 SolderJumper_2_Open
CC1 USB_CC1
JP2 SolderJumper_2_Open
CC2 USB_CC2
NTC Thermistor, OLED Display
+3V3
J4 OLED
OLED_SCL OLED_SDA
TestPoint TP2
THERMISTOR
R6 2.2K
C14 100n
100K@25C
β = 3950
J3 JST-XH
Heat Plate Driver
+12V
C15 2.2u
U6 UCC27511ADBV
IN+ IN− OUTH OUTL GND
PLATE_PWM
R7 22
R8 22
VCC
D4 SS24FL
PLATE_GND
J5 XT30
Q1 DOD130N06
C16 10u
C17 10u
XT30 Connector + Soft Start
VBUS VBUSP
20V − 30V DC
J6 XT30
F1 12A
D5 SMAJ33A
R10 47K
Q2 HSU6119
R13 47K
RC time constant = 47 ms
C18 1u
C19 470u
C20 470u
C21 10u
R11 10K
VCC PWR_FLAG
R9 47K
R12 4K7
C22 100n
VCC_VOLTAGE
User Buttons
+3V3
SW3 SW_SPST R14 10K BTN_L
SW4 SW_SPST R15 10K BTN_R
SW5 SW_SPST R16 10K BTN_T
SW6 SW_SPST R17 10K BTN_B
D6 PESD3V3L5UY
SW7 SW_SPST R20 10K BTN_M
Buzzer
+3V3
BZ1 Buzzer
R18 1K
R19 4K7
BUZZER
Q3 BSS138
R21 100K
M3 Mounting Holes
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
Sheet: /
File: Reflow Plate Controller.kicad_sch
Title: Reflow Plate Controller
Size: A3
Date:
Rev: 1
KiCad E.D.A. 10.0.0
Id: 1/1